

SAVITRIBAI PHULE PUNE UNIVERSITY



A Mini Project Report based on:

SaaS over LAN

Submitted by:

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Under the Support and Guidance of

Prof. A. P. Bhalke

In partial fulfilment of

Cloud Computing Laboratory (310257)

COMPUTER ENGINEERING DEGREE OF

SAVITRIBAI PHULE PUNE UNIVERSITY 2024-2025



CERTIFICATE

This is to certify that following students from **Third Year Computer Engineering** have successfully completed and submitted the **Mini Project** work titled **SaaS Over LAN** at PDEA's College of Engineering, Manjari (BK), Pune – 412307 in the partial fulfillment of the Bachelor's Degree in Computer Engineering by Savitribai Phule University of Pune for the academic year 2024-2025

Ganesh Ghule: (TA-37)

Date:

Guide Name
Prof. A. P. Bhalke

H. O. D.
Dr. M. P. Borawake

ACKNOWLEDGEMENT

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ABSTRACT

This project focuses on the development of a private cloud infrastructure for Software as a Service (SaaS) within a laboratory environment using the existing Local Area Network (LAN). The system is designed with a custom-built cloud controller implemented using open-source technologies, integrated with Hadoop Distributed File System (HDFS) for scalable and fault-tolerant storage. The controller performs essential cloud operations such as segmenting files into blocks, encrypting them for security, and handling file upload and download processes to and from the cloud. This solution ensures data confidentiality, efficient storage, and seamless accessibility, serving as a foundational step toward real-time cloud-based applications in educational or small-scale enterprise environments.

Practical Title:

Setup your own cloud for Software as a Service (SaaS) over the existing LAN in your laboratory. In this assignment you have to write your own code for cloud controller using open-source technologies to implement with HDFS. Implement the basic operations may be like to divide the file in segments/blocks and upload/ download file on/from cloud in encrypted form.

Objectives:

- To set your own cloud for SaaS over existing LAN
- To implement the basic operations may be like to divide the file in segments/blocks

Hardware Requirements:

Pentium IV with latest configuration.

Software Requirements:

Ubuntu 20.04, VMwareESXi cloud

Theory:

Here we are installing VMwareESXi cloud

❓ Host/NodeESXi installation:-

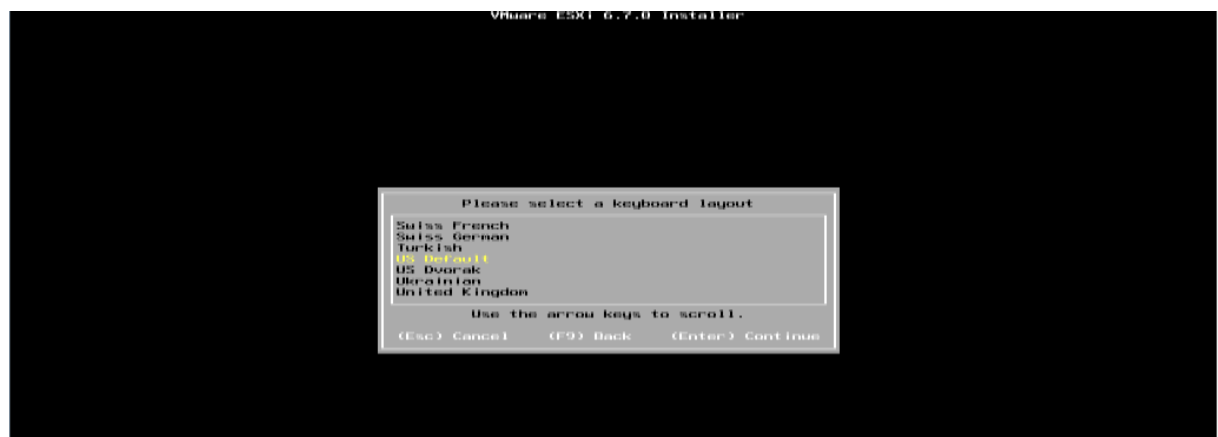
❓ **ESXiHardwareRequirements:-**

- ESXi6.7requiresahostmachinewithatleasttwoCPUcores.
- ESXi6.7supports64-bitx86processors
- ESXi6.7requirestheNX/XDbit to be enabled for the CPU in the BIOS.
- ESXi6.7requiresaminimumof4GBofphysicalRAM.Itisrecommende
d to provide atleast 8 GB of RAM to run virtual machines in
typical productionenvironments.
- Tosupport64-bitvirtualmachines,support for hardware
virtualization (IntelVT-xor AMDRVI)
mustbeenabledonx64CPUs.
- One or more Gigabit or faster Ethernet controllers. For a list
of supportednetwork adapter models.
- SCSI disk oralocal,non-network,RAIDLUN with unpartitioned space

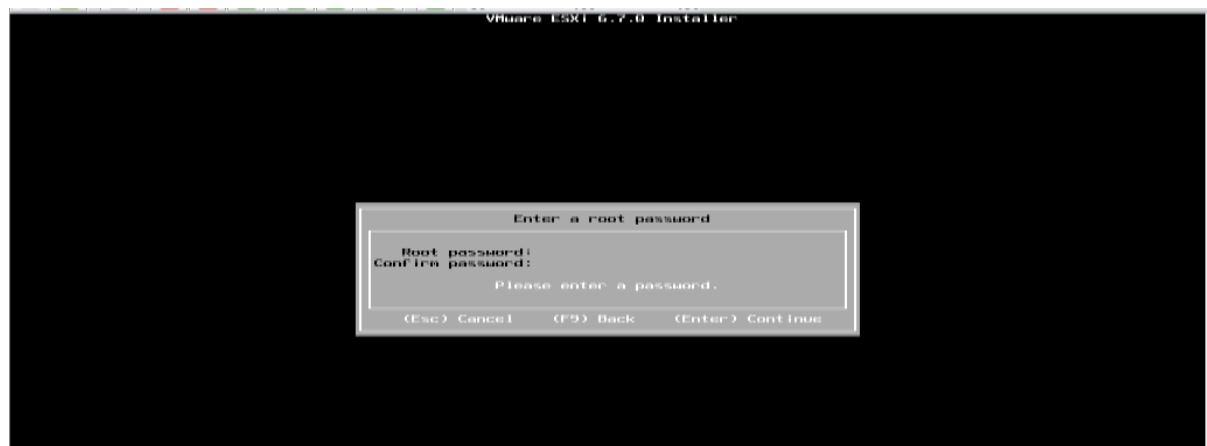
For Serial ATA (SATA), a disk connected through supported SAS controller or supported on board SATA controllers. SATA disks are considered remote not local. These disks are not used as a scratch partition by default because they are seen as remote.



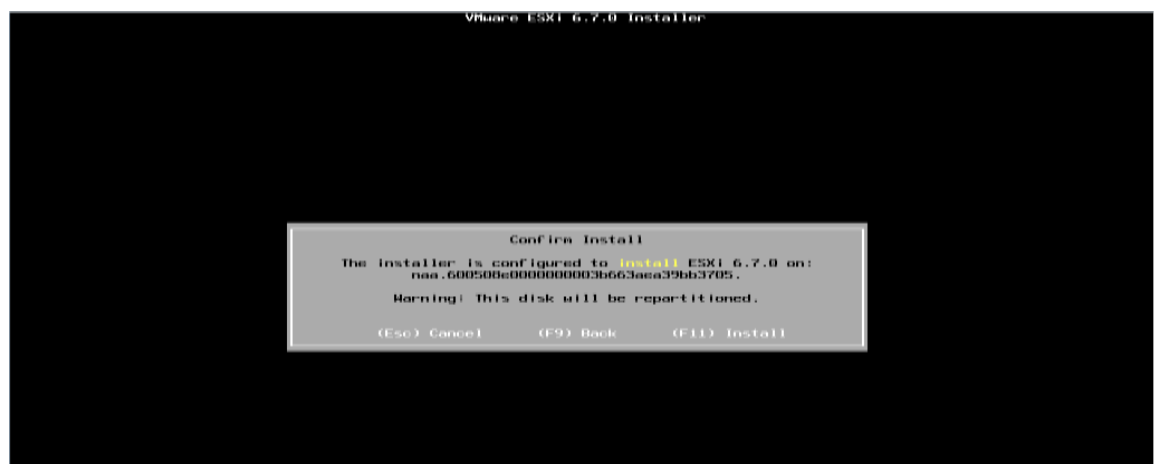
Select storage:



Select Keyboard Layout :



Set NodeESXi Root Password :



Installation complete (Reboot)CLI interface to configuration

System Customization Configure Password Configure Lockdown Mode Configure Management Network Restart Management Network Test Management Network Network Restore Options Configure Keyboard Troubleshooting Options View System Logs View Support Information Reset System Configuration	Configure Password Set To prevent unauthorized access to this system, set the password for the user.
--	---

CLI Interface to Configuration:

Configure Management Network Network Adapters VLAN (optional) IPv4 Configuration IPv6 Configuration DNS Configuration Custom DNS Suffixes	IPv4 Configuration Automatic IPv4 Address: 169.254.157.31 Subnet Mask: 255.255.0.0 Default Gateway: Not set This host can obtain an IPv4 address and other networking parameters automatically if your network includes a DHCP Server. If not, ask your network administrator for the appropriate settings.
---	--

Configure Management Network

Configure Management Network	IPv4 Configuration
Network Adapters VLAN (optional)	Manual
IPv4 Configuration	IPv4 Address: 109.203.103.124 Subnet Mask: 255.255.255.0 Default Gateway: 109.203.103.1
IPv6 Configuration	
DNS Configuration	This host can obtain an IPv4 address and other networking parameters automatically if your network includes a DHCP server. If not, ask your network administrator for the appropriate settings.
Custom DNS Suffixes	

IPv4 Configuration

This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:

☒ **Disable IPv4 configuration for management network**
☐ Use dynamic IPv4 address and network configuration
☐ Set static IPv4 address and network configuration:

IPv4 Address	[109.203.103.124]
Subnet Mask	[255.255.255.0]
Default Gateway	[109.203.103.1]

<Up/Down> Select <Space> Mark Selected <Enter> OK <Esc> Cancel

Configure Management Network	DNS Configuration
Network Adapters VLAN (optional)	Manual
IPv4 Configuration	Primary DNS Server: 0.0.0.0
IPv6 Configuration	Alternate DNS Server: 1.1.1.1
DNS Configuration	Hostname: localhost
Custom DNS Suffixes	If this host is configured using DHCP, DNS server addresses are obtained automatically. If not, ask your network administrator for the appropriate settings.

DNS Configuration

This host can only obtain DNS settings automatically if it also obtains its IP configuration automatically.

☒ **Obtain DNS server addresses and a hostname automatically**
☐ Use the following DNS server addresses and hostname:

Primary DNS Server	[0.0.0.0]
Alternate DNS Server	[1.1.1.1]
Hostname	[localhost]

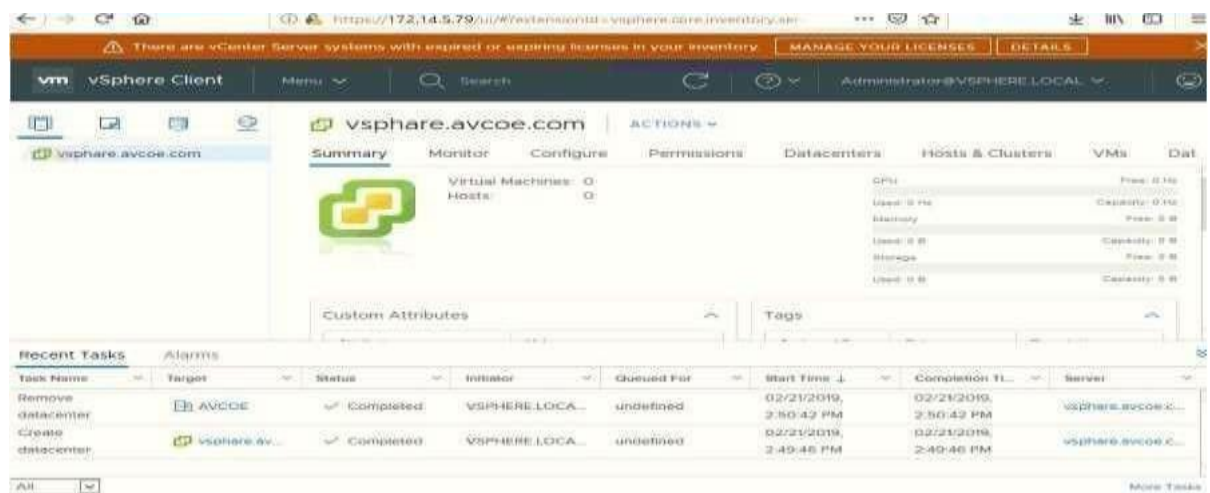
<Up/Down> Select <Space> Mark Selected <Enter> OK <Esc> Cancel

Set DNServer :

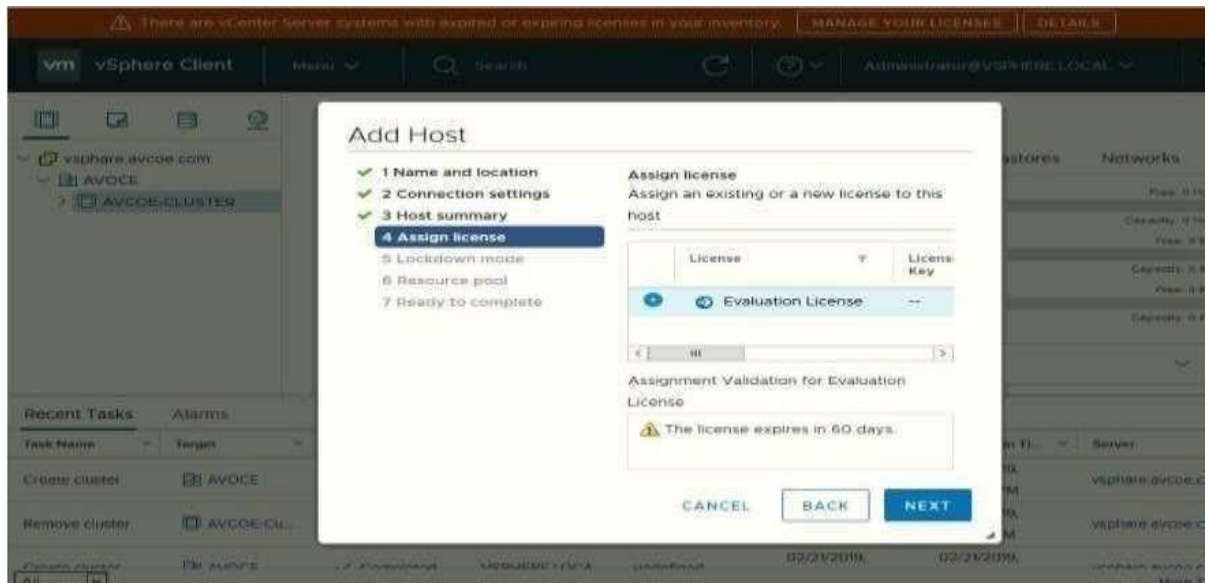
Restart Management Network

System Customization	Restart Management Network
Configure Password Configure Lockdown Mode Configure Management Network Restart Management Network Test Management Network Network Restore Options Configure Keyboard Troubleshooting Options View System Logs View Support Information Reset System Configuration	Restarting the management network interface may be required to restore networking or to renew a DHCP lease. Restarting the management network will result in a brief network outage that may temporarily affect running virtual machines. Note: If a renewed DHCP lease results in a new network identity (e.g., IP address or hostname), remote management software will be disconnected.

GUIAccess :

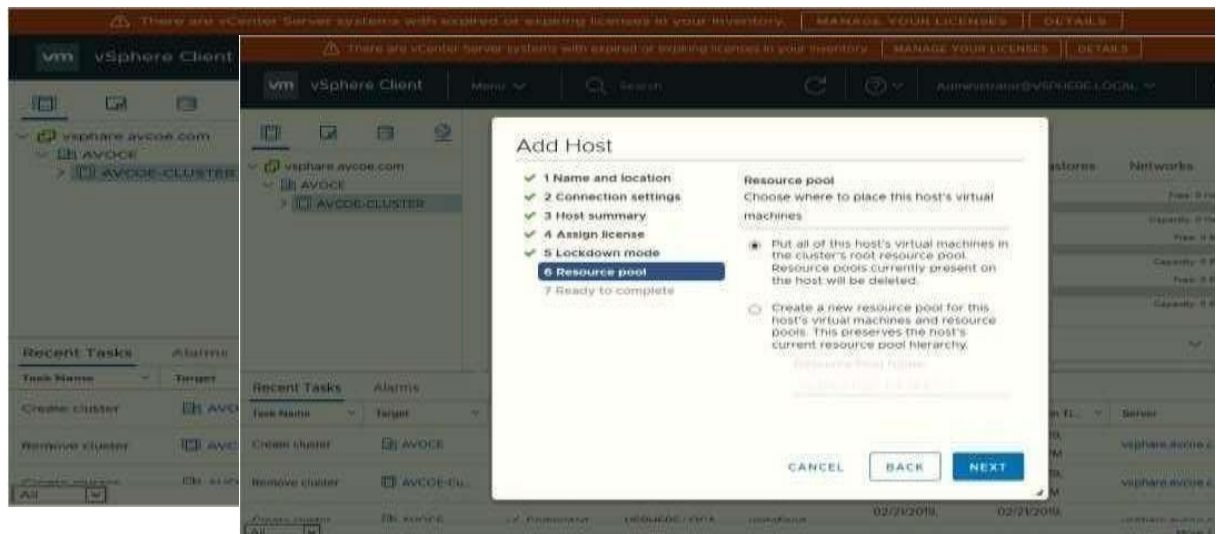


VCenter Access



Lock Down mode:

Add Host In Pool:



Host View and View Config:

Cluster View and Configuration:

There are vCenter Server systems with expired or expiring licenses in your inventory. MANAGE YOUR LICENSES DETAILS

vm vSphere Client

Menu

Search

Administrator@VSPHERE.LOCAL

vsphere.avcoe.com

AVCOE

172.14.5.244

172.14.5.245

172.14.5.245

ACTIONS

Summary

Monitor

Configure

Permissions

VMs

Datstores

Networks

Hypervisor Model: VMware ESXi, 6.7.0, 10302808

Processor Type: Intel(R) Xeon(R) CPU E5-1607 v2 @ 3.00GHz

Logical Processors: 4

NICs: 1

Virtual Machines: 0

State: Connected

Uptime: 0 second

CPU

Free: 11.97 GHz

Used: 0 Hz

Capacity: 11.97 GHz

Memory

Free: 15.95 GB

Used: 0 B

Capacity: 15.95 GB

Storage

Free: 923.58 GB

Used: 1.42 GB

Capacity: 924 GB

Recent Tasks

Alarms

Task Name	Target	Status	Initiator	Queued For	Start Time	Completion TL	Server
Configuring vSphere HA	172.14.5.245	6%	System	156 ms	02/21/2019, 3:04:54 PM		vsphere.avcoe.c...
Add host	AVCOE-CL...	Completed	VSPHERE.LOCA...	undefined	02/21/2019, 3:04:48 PM	02/21/2019, 3:04:54 PM	vsphere.avcoe.c...
Configuring	172.14.5.244	Completed	System	84 ms	02/21/2019,	02/21/2019,	vsphere.avcoe.c...

There are vCenter Server systems with expired or expiring licenses in your inventory. MANAGE YOUR LICENSES DETAILS

vm vSphere Client

Menu

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AVCOE

172.14.5.244

172.14.5.245

AVCOE-CLUSTER

ACTIONS

Summary

Monitor

Configure

Permissions

Hosts

VMs

Datstores

Networks

Total Processors: 8

Total vMotion Migrations: 0

CPU

Free: 21.39 GHz

Used: 2.55 GHz

Capacity: 23.94 GHz

Memory

Free: 29.9 GB

Used: 2.96 GB

Capacity: 31.85 GB

Storage

Free: 1.8 TB

Used: 2.84 GB

Capacity: 1.8 TB

Related Objects

vSphere DRS

Recent Tasks

Alarms

Task Name	Target	Status	Initiator	Queued For	Start Time	Completion TL	Server
Configuring vSphere HA	172.14.5.245	Completed	System	156 ms	02/21/2019, 3:04:54 PM	02/21/2019, 3:05:34 PM	vsphere.avcoe.c...
Add host	AVCOE-CL...	Completed	VSPHERE.LOCA...	undefined	02/21/2019, 3:04:48 PM	02/21/2019, 3:04:54 PM	vsphere.avcoe.c...
Configuring	172.14.5.244	Completed	System	84 ms	02/21/2019,	02/21/2019,	vsphere.avcoe.c...

All

More Tasks

Conclusion:

In this project, we successfully implemented a private cloud infrastructure within our laboratory environment using open-source technologies and integrated it with HDFS for reliable and distributed storage. By writing custom code for a cloud controller, we were able to simulate the basic functionalities of a Software as a Service (SaaS) model, including segmenting large files into blocks, encrypting them for secure transfer, and enabling upload and download operations across the local cloud setup.

This practical exercise gave us hands-on experience with cloud computing fundamentals, data security practices, and distributed file systems. It helped us understand the role of cloud controllers and storage nodes in managing file transfers, encryption mechanisms, and system performance over a LAN setup.

Additionally, we explored the configuration and management of a **vSphere Private Cloud**, which allowed us to understand how enterprise-level virtualization platforms operate. Through this, we learned about resource allocation, VM provisioning, and cloud service delivery in a controlled private network environment.

Overall, this assignment strengthened our practical knowledge in cloud computing and provided a solid foundation to explore more advanced cloud services and architectures in the future. It also enhanced our understanding of how cloud-based SaaS models can be implemented securely and efficiently in both educational and enterprise settings.